Engine Model(s): PT6A

72-50-02

COMPRESSOR TURBINE - DESCRIPTION AND OPERATION

1. Description and Operation (Ref. 72-50-00, Fig. 1)

The compressor turbine installation consists of a two-plane balanced turbine disk with an integral shaft extension, turbine blades, and classified weights. The assembly is an independently balanced unit which can be replaced by another balanced unit without affecting the compressor rotor assembly balance.

The turbine drives the compressor rotor assembly in a counterclockwise direction through the shaft extension, which is externally splined to fit into the internally splined compressor stubshaft. A master spline ensures positive location of the turbine assembly to retain original balance. A center bolt and keywasher secure the turbine assembly to the compressor rotor assembly.

The turbine disk embodies a circumferential groove to enable disk growth to be checked. The blades are secured to fir-tree serrations machined in the disk circumference and retained in position by rivets. Blades are of cast high temperature alloy and are parallel or tapered according to requirements of individual engine model. They incorporate squealer tips to make sure minimum rub should they contact the shroud segments. Classified balance weights are riveted, as necessary, to the front and rear disk flanges during turbine balancing

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